

INFOGRAPHIC: Understanding Likelihood Theory

The Backbone of Statistical Modelling

[PROBABILITY]

Given parameters (θ) $\rightarrow$ Predicts Data (X)

[LIKELIHOOD]

Given observed Data (X) $\rightarrow$ Estimates (θ)

1. The Core Concept: Probability vs. Likelihood

While they use the same mathematical function, they look at the world from opposite directions.

- **Probability:** You know the coin is fair ($p = 0.5$). What is the chance of getting 3 heads?
Formula: $P(\text{Data} \mid \theta)$
- **Likelihood:** You flipped a coin 10 times and got 8 heads. How likely is it that the coin is fair?
Formula: $L(\theta \mid \text{Data})$

Key Takeaway: Probability calculates the plausibility of *future data* given known parameters. Likelihood evaluates the plausibility of *different parameter values* given already observed data.

2. The Likelihood Function: $L(\theta \mid X)$

Imagine you collect a dataset of independent observations $X = \{x_1, x_2, \dots, x_n\}$.

The Math Behind the Curve

Because the data points are independent, we multiply their individual probabilities:

$$L(\theta \mid X) = \prod_{i=1}^n f(x_i \mid \theta)$$

- **The Problem:** Multiplying many small probabilities results in tiny numbers (underflow issues), and differentiating products is hard.
- **The Solution:** The **Log-Likelihood**, denoted as $\ell(\theta)$.

$$\ell(\theta) = \ln L(\theta \mid X) = \sum_{i=1}^n \ln f(x_i \mid \theta)$$

Turning products into sums makes calculus much easier!

3. Maximum Likelihood Estimation (MLE)

The goal of statistical modelling is to find the parameter $\hat{\mathbf{i}}(\theta)$ that maximizes our Log-Likelihood function. In other words, we want to find the parameter value that makes our observed data "most expected".

The 3-Step Recipe for MLE

1. **Write** the Log-Likelihood function $\ell(\theta)$.
2. **Take the derivative** with respect to θ (this is called the **Score Function**, $U(\theta)$):

$$U(\theta) = \partial \ell(\theta) / \partial \theta$$

3. **Set to zero** and solve for θ :

$$U(\hat{\mathbf{i}}_{MLE}) = 0$$

4. Measuring Uncertainty: Fisher Information

How confident are we in our estimate $\hat{\mathbf{i}}(\theta)$? Look at the curvature of the log-likelihood peak.

Sharp Peak (High Curvature)	Flat Peak (Low Curvature)
High Fisher Information	Low Fisher Information
Data gives a strong, precise signal	Data is noisy; many parameters are plausible
Low variance/uncertainty in $\hat{\mathbf{i}}(\theta)$	High variance/uncertainty in $\hat{\mathbf{i}}(\theta)$

Observed Fisher Information: The negative second derivative at the MLE:

$$I(\hat{\mathbf{i}}_{MLE}) = - [\partial^2 \ell(\theta) / \partial \theta^2] |_{\theta=\hat{\mathbf{i}}_{MLE}}$$

5. Hypothesis Testing with Likelihood

Once you have your model, how do you test if a specific parameter value is correct? We use the **Likelihood Ratio Test (LRT)**.

$$\lambda = L(\text{Null Model} / \text{Restricted}) / L(\text{Alternative Model} / \text{Unrestricted})$$

- **The Logic:** We compare the maximum likelihood of a simple model (H_0) against a more complex model (H_1).
- **The Test:** If the complex model fits *significantly* better, the ratio drops, and we reject H_0 .

- **Wilks' Theorem:** For large samples, $-2 \ln(\lambda)$ follows a Chi-squared (χ^2) distribution.